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Abstract

1 Introduction

Lake ice in the Great Lakes is extremely variable and has impacts ranging from economy to ecology. The commercial shipping industry in the Great Lakes transports 135 million tons of cargo each year and supports a quarter million jobs (Martin, 2023). The presence vs absence and timing of ice cover influences the length of the shipping season and during high ice years shipping has been entirely suspended which interrupts supply chains that rely on raw materials such as Iron Ore, Grain, etc. The biota of the lakes also interact with ice cover which impacts nutrient loading (Nicholls, 1998), serves as a winter habitat for Diatoms (Twiss et al., 2012) and has been widely suggested to protect Lake Whitefish eggs during otherwise harsh winter conditions (Brown et al., 1993; Collingsworth et al., 2017; Lynch et al., 2015).

Previous studies have demonstrated that, consistent with trends in air and ocean temperatures, water temperatures in the Great Lakes, are also increasing both in the surface (Austin and Colman, 2007, many more!) and subsurface Anderson et al., 2021. However, the rate and extent of warming in the Great Lakes varies considerably from lake to lake and even *within* each lake (Mason et al., 2016), partly due to the heterogeneity of characteristics within lakes (bathymetry, latitude,) often referred to as lake morphometry. Richardson et al. (2024) hypothesized that, globally, lake ice phenology will respond nonlinearly to climate-change such that lakes in geographically colder (warmer) regions experience increased (decreased) interannual variability. This is fairly consistent with the findings of Lin et al. (2022), who showed that regionally the Great Lakes have been experiencing increased variability since 1998.

2 Data & Methods

Ice cover in the Great Lakes has been recorded for over five decades with information from high resolution, satellite-based instruments.

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While much previous work has identified the significant influence of teleconnections on (Great) Lakes ice cover (Bai et al., 2012; Richardson et al., 2024; Wang et al., 2018), our focus is on subregional ice formation (i.e. differences